The MBT Sustainable Cycle: From Net Zero to Bio-Positive

A comprehensive framework for Earth System regeneration via Microbial BioTransformation (MBT55).

Redesigning the flow of electrons to bridge the Metabolic Rift.

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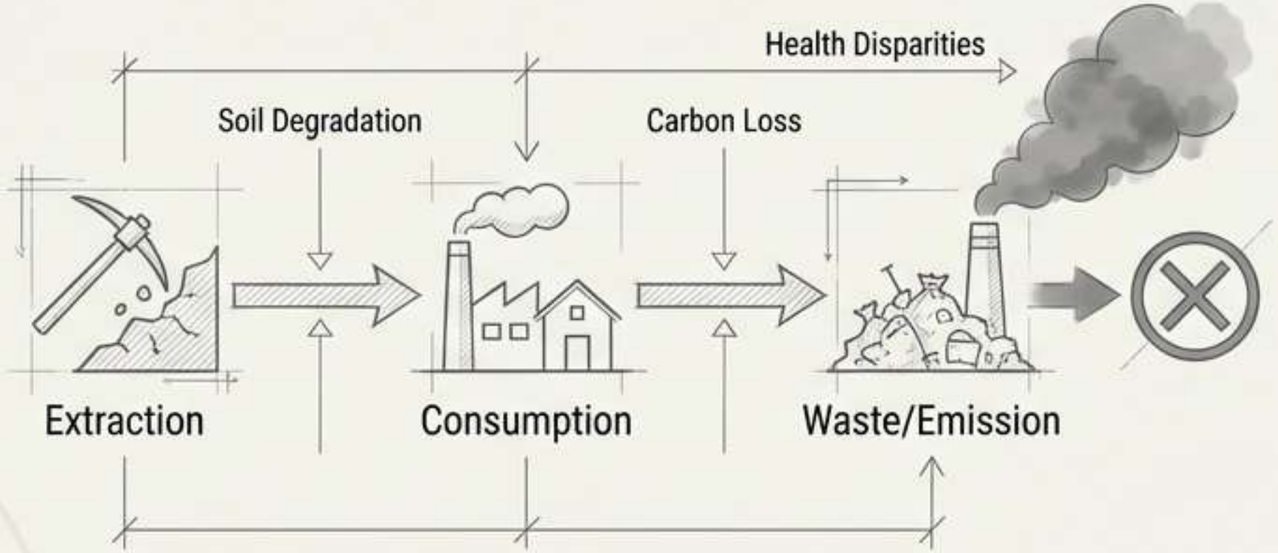
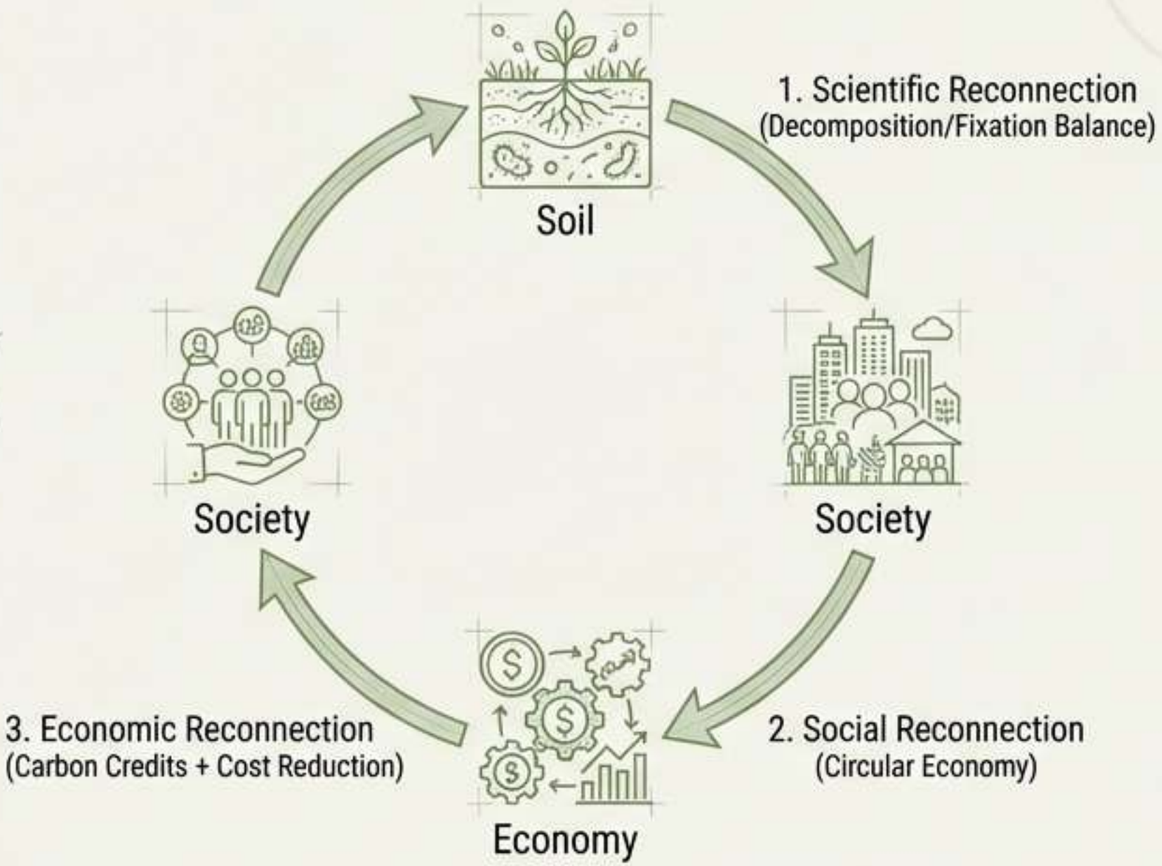
The Metabolic Crisis: A System Disconnected

Suisse Int'l The Metabolic Rift (Current)



The Reconnected System (MBT55) Suisse Int'l



Suisse Int'l: The Opportunity: Moving beyond fragmented Carbon Removal to Ecological Hyper-cycles

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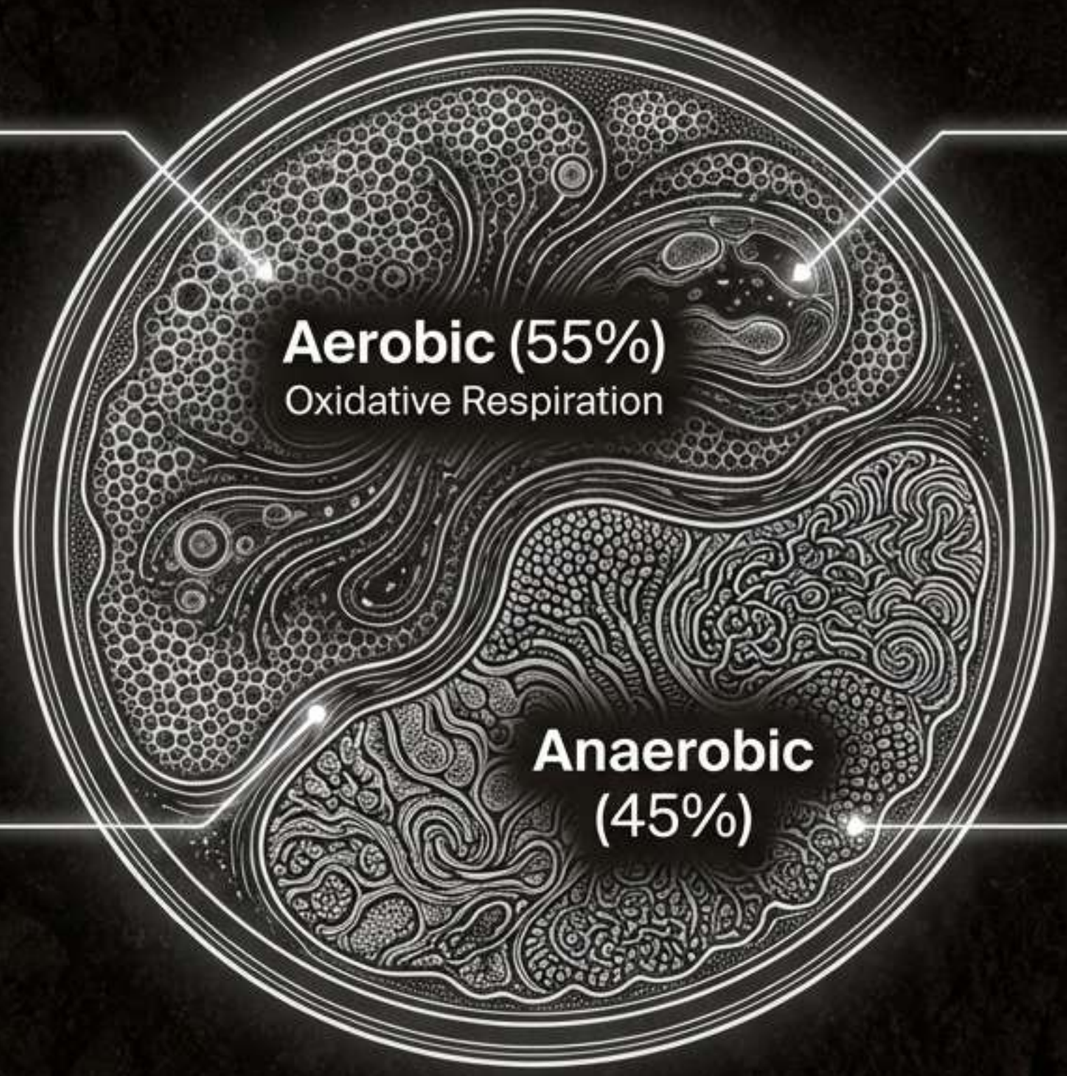
The Engine: MBT55 Microbial Consortium

Carbon Decomposition: Converting recalcitrant organics (cellulose, lipids) into energy.

N-Cycle: Nitrogen fixation and ammonia oxidation.



Redox Mediation: Electron transfer maintenance.

Biofilm Network: Membrane structure for substance/information transfer.

Differentiation: Unlike compost or biochar, MBT55 achieves Simultaneity of Decomposition = Generation.

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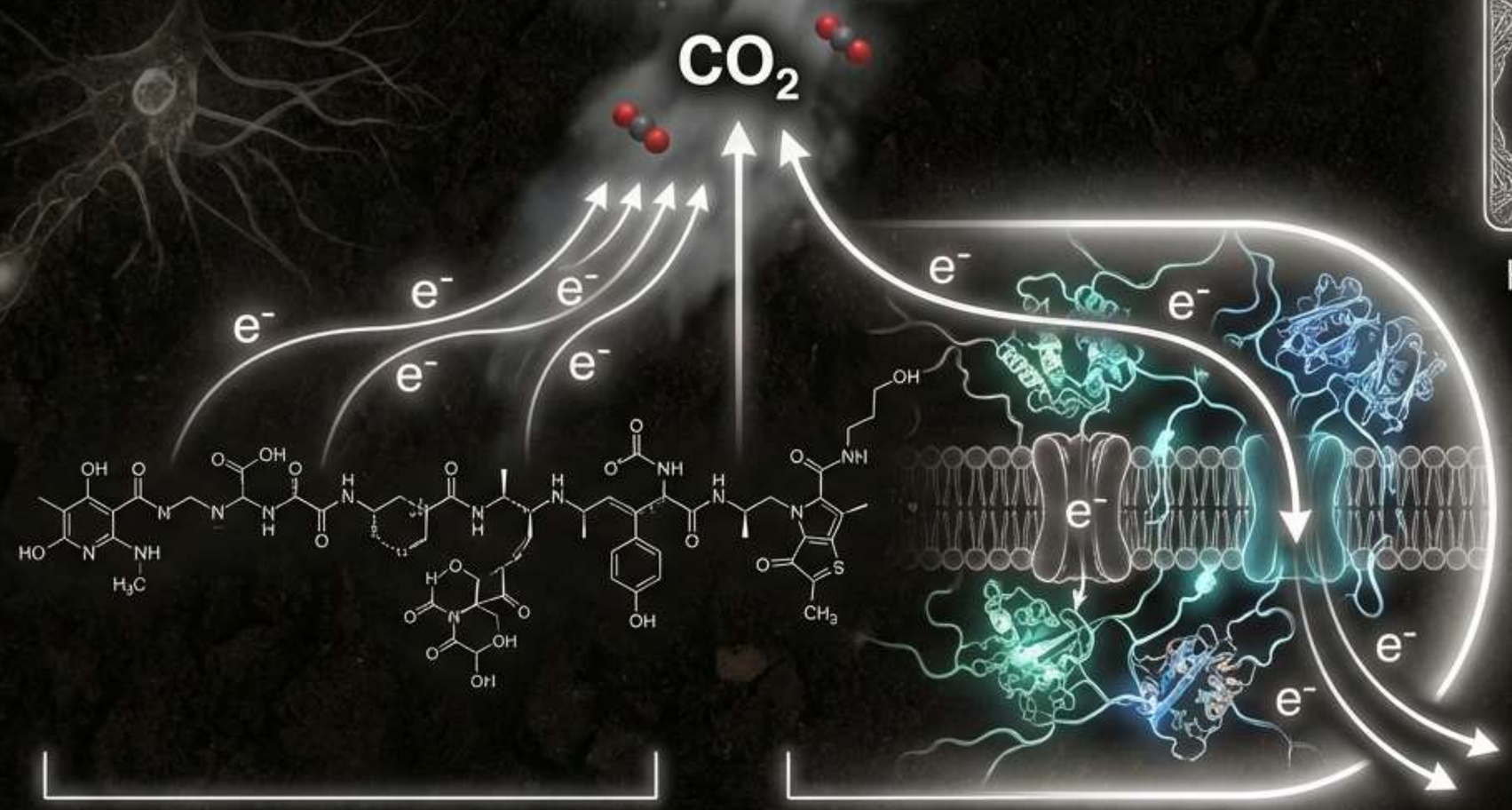
Redesigning Electron Flow

The Mechanism: Reconnecting Electron Flow to Nature



$$C_f = \lambda \cdot \frac{E_d}{E_d + K_m}$$

Higher Electron Donor Efficiency (E_d) drives the fixation rate (C_f).

Stable Carbon Fixation
(>100-year sequestration)

"Carbon regeneration through electron flow reconnection."

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Beyond Conventional Regenerative Tech

Positioning MBT55 as an Ecological Engineering Platform

Technology	Carbon Fixation Efficiency	Microbial Diversity	Economic Viability
Conventional Compost	★	★★	★★★
Biochar	★★☆	★	★★
Chemical Fertilizer	☆	☆	★★★☆
MBT55	★★★★★	★★★★★	★★★★★

Key Insight: MBT55 integrates biological function with chemical stability. It is not just a composting technology, but an ecological engineering platform that redesigns the carbon cycle.

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The M³-BioSynergy System (MSC)

Layer 3: Economic (Eco-Financial).

Value by Circulation.
Carbon Credits, Green Premiums.

Layer 2: Information (Bio-Informatic).

Distributed Data Assimilation.
Bio-Knowledge Graph via Azure AI.

Layer 1: Material (Biophysical).

Waste to Resource.
SOC Formation & Methane Reduction.

The Eco-Hypercycle: Where material, information, and economy resonate in phase.

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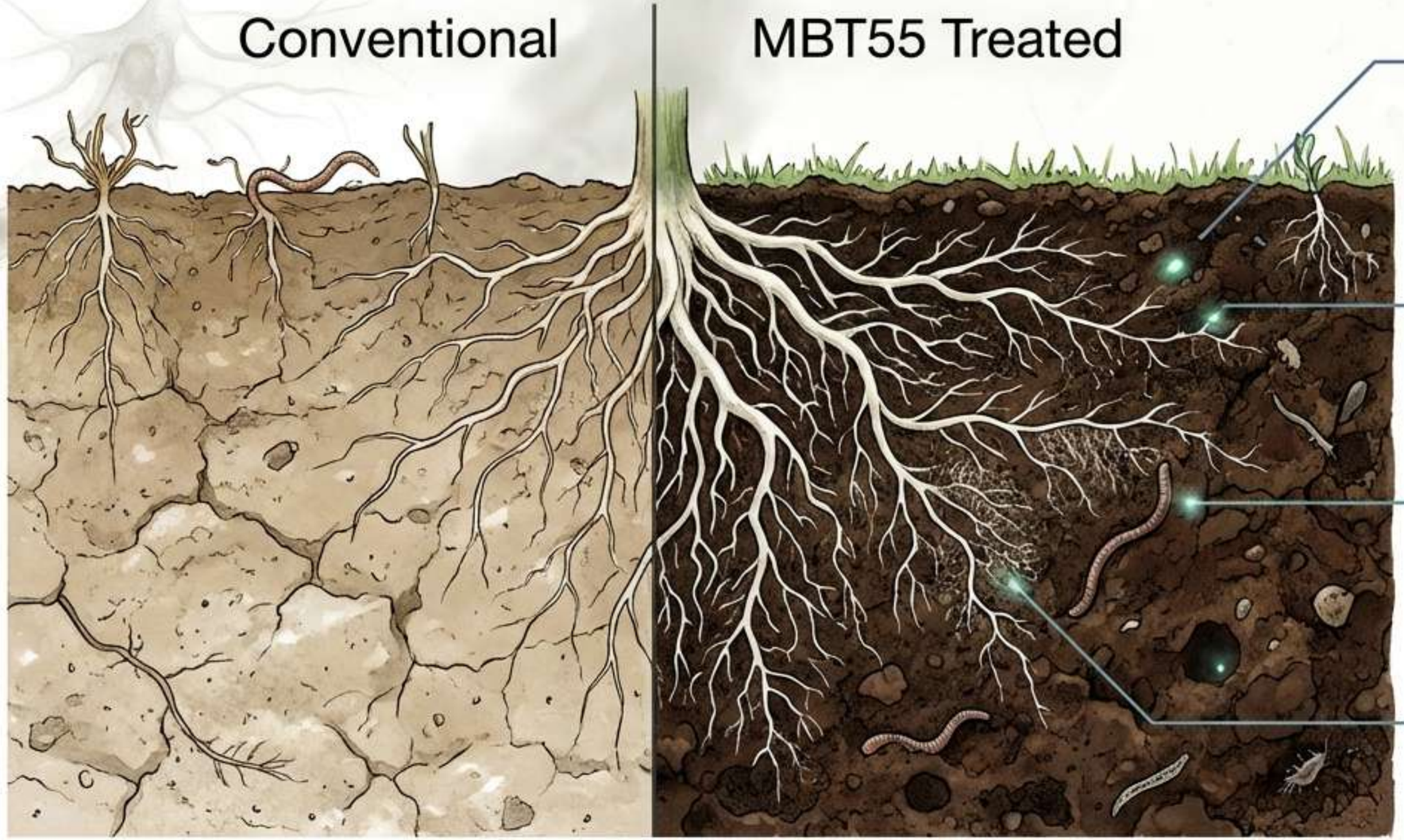
Quantitative Recovery of Earth Systems

Nairobi/Kenya Pilot Data



+116%
SOC Formation Rate
(0.6% -> 1.3%)

2.4 t/ha/year
CO_2 Sequestration

+21%
Water Retention

-48%
Ammonia Volatilization

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The Micro-Macro Link: Soil to Gut

Health & Nutrition Impact

Mechanism

- MBT-grown crops -> Higher minerals/nutrients.
- Product Spotlight: reCLA (MBT Food Probiotics). Metabolites improve gut microbiome.

Impact

- Gut environment improvement: +32–48%
- Medical cost reduction: ~$120 per person/year
- Platform: Integration with HealthBook for risk assessment.

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Creating a Circular Intelligence Economy

Body and data text uses Suisse Int'l

The Formula:
$V_{gp} = (C_{red} + N_{fix}) \times P_c$

GDP Impact:
+2.8% to +3.4% annual contribution.

Multiplier Effect:
1.72 (Logistics, Energy, Education).

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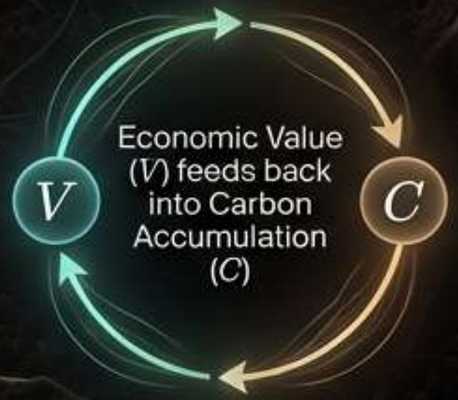
The Mathematics of Sustainability

MSC Sustainability Index (SI)

Suisse Int'l

$$SI = w_1 E + w_2 S + w_3 Ec$$



E (Environmental): 0.40

40%

S (Social): 0.35

35%

Ec (Economic): 0.25

25%

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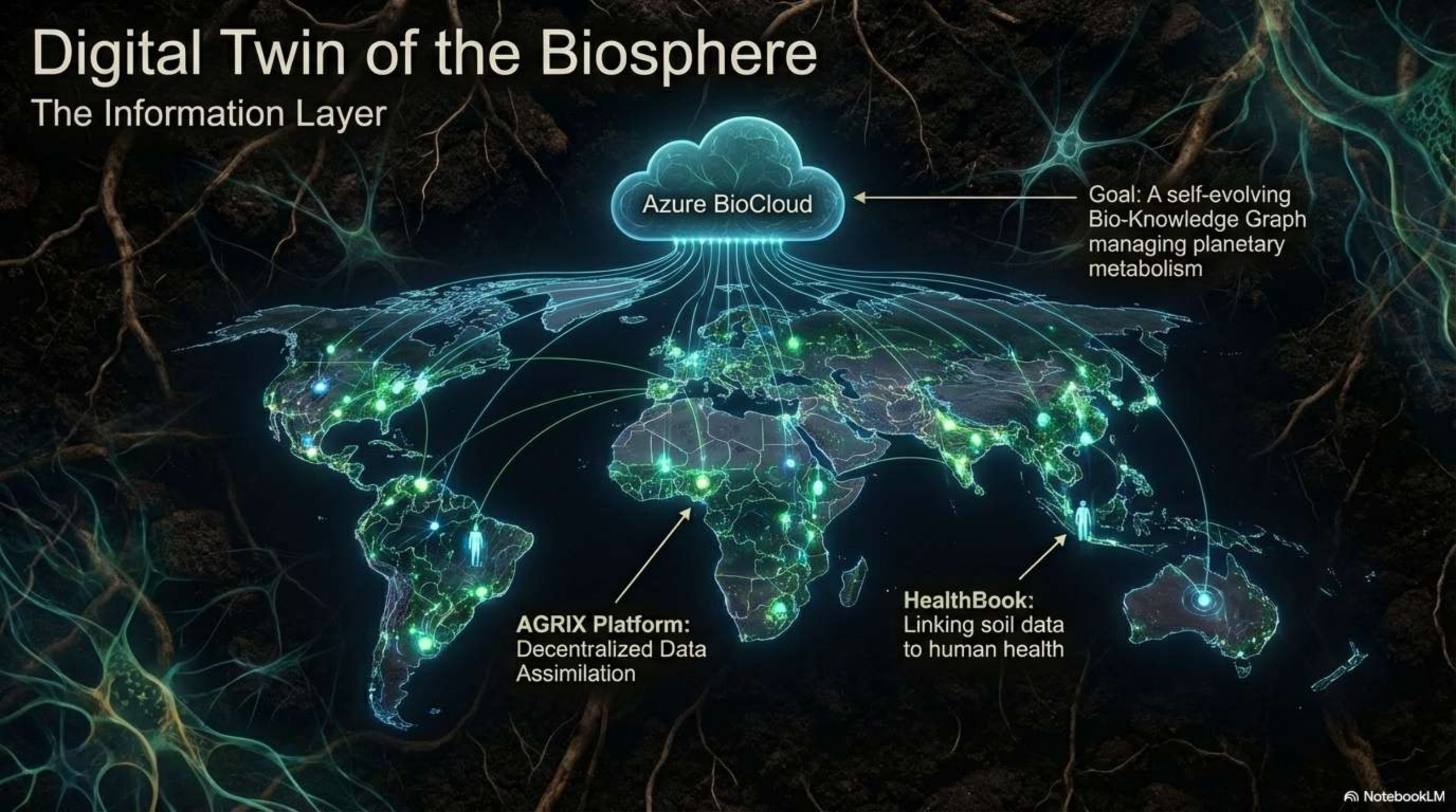
Digital Twin of the Biosphere

The Information Layer



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Global Implementation Roadmap (2025–2050)

2025-26: Phase 1.
Suisse Int'l
Scientific Foundation.
Kenya Pilot.

2026-28: Phase 2.
Suisse Int'l
Pan-African Pilot.
20k ha target.

2028-30: Phase 3.
Editorial New
Financial Productization.
Carbon Loop Bonds.

2030-35: Phase 4.
Editorial New
Decentralized AI.
Earth Microbial Brain.

2035-50: Phase 5.
Editorial New
Global BioNexus.
Carbon Negative Society.

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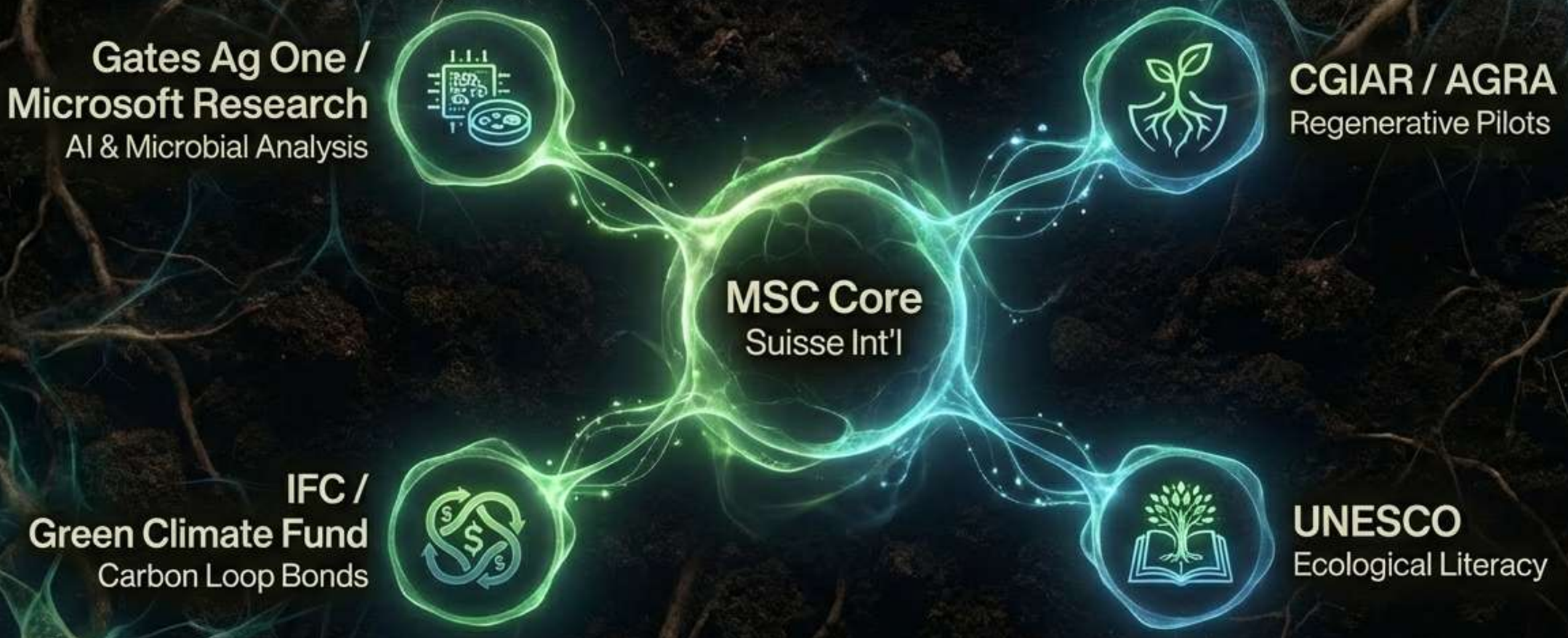
A Strategic Alliance for Agri-Regeneration



Proposal: Scientific Research Grants, Social Implementation, & Environmental Finance

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Co-Evolutionary Governance (in Editorial New)

- Philosophy: Shifting from Anthropocentrism to Life-Centrism.
- New Economy Principles:
  1. Nature as Infrastructure.
  2. Value by Circulation.
  3. Recursive Ecosystem: Humans, Microbes, and AI.

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From Carbon Neutral
to Carbon Intelligence
(in Editorial New)
MBT Sustainable Cycle is not just a solution,
but a new civilization model.
Join the transition to a Bio-Positive Earth.
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